

Q1. Define Worms and Viruses

Computer Virus:

A computer virus is a malicious program that attaches to a host file and spreads when the file is executed, often damaging data.

Worm:

A worm is a self-replicating program that spreads automatically across networks without user action and consumes system resources.

Q2. Types of Malicious Programs (with one-line description)

- **Virus** – Attaches to files and spreads when executed.
 - **Worm** – Self-replicates and spreads over networks automatically.
 - **Trojan Horse** – Malicious program disguised as legitimate software.
 - **Ransomware** – Encrypts data and demands payment for access.
 - **Spyware** – Secretly collects user information.
 - **Adware** – Displays unwanted advertisements.
 - **Rootkit** – Hides malicious activities from detection.
 - **Keylogger** – Records keystrokes to steal sensitive data.
 - **Botnet** – Network of infected systems controlled remotely.
 - **Logic Bomb** – Activates when a specific condition is met.
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Q3. Types of Computer Viruses (with one-line description)

- **Boot Sector Virus** – Infects the boot sector and runs at system startup.
 - **File Infector Virus** – Attaches to executable files and spreads when opened.
 - **Macro Virus** – Infects documents using macros (e.g., Word/Excel files).
 - **Multipartite Virus** – Infects both boot sector and files.
 - **Polymorphic Virus** – Changes its code to avoid antivirus detection.
 - **Stealth Virus** – Hides its presence from the system and antivirus.
 - **Resident Virus** – Stays in memory and infects files continuously.
 - **Non-Resident Virus** – Infects files only when executed and then becomes inactive.
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Q4.Buffer Overflow (Short Note)

A **buffer overflow** is a condition in which a program writes more data into a memory buffer than it can hold, causing the extra data to overwrite adjacent memory locations.

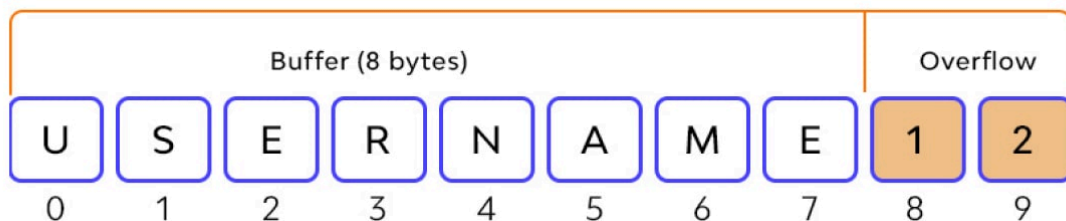
- It occurs due to **lack of proper bounds checking** in programs.
- Attackers can exploit it to **execute malicious code** or crash the system.
- Common in languages like **C and C++** that do not perform automatic memory checks.
- It may lead to **system crashes, data corruption, or unauthorized access**.

Prevention Methods:

- Use **bounds checking**
- Apply **secure coding practices**
- Use **modern programming languages** (Java, Python)
- Implement **stack canaries and memory protection techniques**



Buffer overflow example



Types of Buffer Overflow Attacks:

- **Stack-based Buffer Overflow:**

This happens when the overflow occurs on the stack. It can overwrite the return address and other control data, allowing the attacker to control the flow of execution.

- **Heap-based Buffer Overflow:**

This type of overflow happens in the heap. It can corrupt function pointers or other critical data structures used by the program.

- **Format String Vulnerabilities:**

A type of overflow where an attacker can manipulate format strings (e.g., %s or %x) to leak information or overwrite memory.